Retail CCTV Footfall Tracking & Live Occupancy System

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1. PIPELINE ARCHITECTURE & METHODOLOGY

- **Human Detection (YOLOv8)**: Executes inference (τ_{conf} = 0.35, NMS IoU = 0.50) targeting person instances (COCO Class 0) at native CCTV resolution with real-time throughput.
- **Multi-Object Tracking (ByteTrack)**: Manages spatial association via Kalman state forecasting and two-stage Hungarian assignment, retaining low-confidence detections during occlusion.
- **Ground-Plane Geometric Tripwire**: Evaluates bottom-center foot contact points (x_{center}, y_2) across a directed vector across the entrance doorway mat (0.55W, 0.85H → 0.98W, 0.72H). Anchoring on the feet prevents perspective parallax and body-tilt errors.
- **Directional State Machine**: Evaluates signed perpendicular distance across the vector. Crossing Outside → Inside increments total IN and live headcount; Inside → Outside increments OUT and decrements headcount.

2. HANDLING CRITICAL REAL-WORLD EDGE CASES

Edge Case	Core Vulnerability	Implemented Solution
Occlusion	Display fixtures, shelves, and passing shoppers obscure bounding boxes, dropping track continuity.	ByteTrack associates low-score detections (0.10 ≤ score < 0.35) with predicted Kalman states; tracklets survive up to 30 frames of total visual occlusion.
Groups Entering	Shoppers entering side-by-side produce heavily intersecting bounding boxes.	Foot-point coordinates on the floor plane remain spatially distinct even when upper bodies overlap. Independent per-ID state machines record simultaneous crossings.
Re-entering Shoppers	Customers lingering on the threshold or stepping out briefly trigger duplicate counts.	A spatial hysteresis band (±18px) and a 1.5s temporal debounce prevent oscillation. For prolonged exits, cosine similarity on OSNet Re-ID embeddings links visits.
Staff vs. Customers	Cashiers and stocking clerks remain in-store, skewing dwell times and tripwire counts.	A static spatial exclusion mask covers the cash register zone. Staff exhibit closed-loop aisle trajectories, unlike linear customer transit paths.

3. FAILURE MODES & PRACTICAL ENGINEERING FIXES

- **Threshold Hovering / Sign Flipping:** Customers pausing on the threshold mat caused bounding-box micro-jitter, flipping the signed distance and logging false rapid events. *Resolution:* Implemented a dual-boundary hysteresis deadband (±18px) and a frame-based cooldown debounce window.
- **Perspective Scale Variance:** People near the camera entrance appear large (~450px) while people in rear aisles appear small (~70px), distorting velocity. *Resolution:* Positioned tripwire strictly on the entrance ground-mat where scale and distance to camera are constant.
- **Cold-Start Store Population:** Starting tracking at headcount = 0 in an operating store led to negative occupancy when initial shoppers exited. *Resolution:* Parameterized baseline offset (--initial-headcount 6) calibrated from initial frame audit.

4. FUTURE ENGINEERING ROADMAP

- **Planar Homography / Bird's-Eye View (BEV)**: Compute a 4-point perspective warp matrix to project foot coordinates onto real-world metric store coordinates (meters), enabling true walking speed and customer density metrics.
- **Deep Visual Re-Identification (Re-ID)**: Integrate a lightweight OSNet/FastReID backbone to compute 512D feature embeddings, preserving customer identities across multi-hour intervals and multi-camera handover.
- **Multi-Zone Retail Analytics**: Transition to multi-polygon spatial state machines (Entrance Foyer, Grocery Aisles, Checkout Queue) to compute dwell-time distributions, product attraction heatmaps, and checkout queue wait times.
- **Staff Identification via Vision-Language**: Deploy a zero-shot CLIP or HSV color histogram classifier to detect corporate apron/uniform colors, segmenting employee activity automatically without manual spatial masks.